ORBCOMM®

CONNECTING THE WORLD'S ASSETS



GT 12xx
Hardware Guide

GT0023, Version 06
Nov 2022

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1 OVERVIEW

ORBCOMM's solar-powered, IP67 rated, GT 1200, GT 1210, GT 1220, and GT 1230 (hereinafter referred to as GT 12xx unless noted otherwise) deliver the latest in tracking technology for dry trailer, container, vessel, and other transport assets.

A solar panel leverages the latest technology to deliver long-lasting battery life and charging capabilities, making recharging possible even in indirect light, on overcast days and in high Northern latitudes. Shipped fully charged, the devices hold their charge for at least 6 months, allowing for storage prior to installment. Once installed, the device can deliver multiple messages per day with limited sun exposure.

Device upgrades are possible either over-the-air or using the serial port. Although configurable, the device ships with a standard set-up for ping rates and start/stop reporting.

ORBCOMM offers the following models and configurations:



- GT 1200 / GT 1220

  Get complete visibility for trailers and containers without maintenance or battery changes with advanced reporting and analytics to help streamline processes and improve efficiencies.

- GT 1210 / GT 1230

  This all-in-one, feature-rich tracking device with integrated cargo sensor delivers powerful remote asset management and cargo status detection capabilities with minimum wiring.

1.1 Web Platform

- ORBCOMM Platform is a single, unified, next-generation analytics and reporting cloud-based platform that puts everything in one place with intuitive icons to distinguish asset types, reports by asset type and customizable workspaces and access control so that each user gets the data they need quickly and easily.
- ORBCOMM DeviceCloud (ODC) is a single interface for managing multiple networks and devices, where connectivity and device-specific messaging is abstracted to a common interface and messaging API. ODC allows you to perform tasks like changing the default GT 12xx reporting interval or checking the current GT 12xx battery voltage.

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- CargoWatch® is a web application that delivers near-real-time alerts on trailer status, location, history, as well as arrival and departure, providing greater visibility and utilization for fleet managers. CargoWatch allows you to perform tasks such as checking the current GT 12xx battery voltage.

1.2 Bluetooth®

The Bluetooth (BLE) wireless module in each GT 12xx allows wireless communication with select ORBCOMM sensors . Bluetooth wireless technology allows dual-mode devices to share a single radio antenna.

1.3 ORBCOMM Field Support Tool

The Field Support Tool (FST) is a mobile app (“the app”) that assists in the installation, activation, and maintenance of various ORBCOMM products such as GT 12xx devices and associated accessories. It is primarily used to identify assets and pair wireless accessories to ORBCOMM devices.

This app operates on mobile phones and tablets using Bluetooth wireless technology to communicate with ORBCOMM devices and accessories such as the ORBCOMM Wireless Door Sensor (DS 300).

Refer to document *MA002 Field Support Tool User Guide* for more details.

1.4 Sensors

Add powerful ORBCOMM sensors for added visibility and reporting (APPENDIX B).

Contact your ORBCOMM Account Executive to discuss the features and options available to suit your business requirements.

1.5 Installation

This product was specifically designed for flexible, quick and easy installation. It is installed using VHB tape, and rivets, mounted vertically or horizontally, on containers or trailers, including sheet and post. As it includes integrated antennas, cargo and Bluetooth sensors, no wiring is necessary for most applications. It is sized to fit container corrugations or trailer roof mounts, keeping the device protected as much as possible.

Refer to the various GT 12xx installation guides available from ORBCOMM Customer Care or the Technical Publications download website.

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1.6 Features and Models

The GT 12xx is available in a variety of models, with or without a connector. Contact your ORBCOMM Account Executive for other models and options.

Table 1: Configurations

Features	Models			
	GT 1200 / GT 1220 with Connector	**GT 1200 / GT 1220 without Connector**	**GT 1210 / GT 1230 with Connector**	**GT 1210 / GT 1230 without Connector**
LTE/3G/2G	YES			
Global LTE	YES			
Solar Charge	YES			
Internal Battery	YES			
16-Pin Connector	YES	NO	YES	NO
Internal BLE	YES			
Tamper Detection	YES			
Tracking Mount Height - 1.2m (4 feet)	NO	NO	YES	YES
Integrated Cargo Sensor	NO	NO	YES	YES

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2 SPECIFICATIONS

2.1 Temperature

Parameter	Value
Operating Temperature Range excluding battery charging	-40°C to +70°C (-40°F to +158°F)
Storage Temperature Range	-40°C to +60°C (-40°F to +140°F)

2.1.1 Backup Battery Controller Temperature

Table 2 defines the backup battery's temperature specifications.

Table 2: Backup Battery Temperature Specifications

Parameter	Value
Charge Temperature Range	-20°C to +60°C (-4°F to +140°F)

2.2 Electrical

2.2.1 Input Range

Parameter	Value
Power Supply Voltage	9 to 32 V DC

2.2.2 Power Consumption

Typical values for a nominal battery voltage of 7.4 VDC. The maximum current is 1100 mA.

Table 3: Input Currents

Mode of Operation	Current	Unit
Sleep[1] + Power Save	150	µA
Cellular Transmit (pulsed)	1000	mA
Cellular Receive	45	mA
Application Processing	25	mA

2.2.3 Maximum Input Power

The maximum input power required for the battery charge is 3.95 W.

2.2.4 Load Dump Protection

The configurable I/O is protected for load dump.

[1]Sleep current is based on current hardware at room temperature.

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2.2.5 Reverse Voltage Input

Parameter	Voltage
Reverse Polarity Protection	-40 V DC (maximum)

2.2.6 Solar Panel

The solar panel, built-in to each GT 12xx, supplies power for recharging the device's internal battery. If completely discharged, it can be placed in direct sunlight to recharge.

The solar panel electrical specifications are below.

Parameter	Min.	Typical	Max.	Units
Voltage	10.5	-	-	V
Power	-	2.5	-	W

2.3 Connectors

2.3.1 Connector Pin Assignment

Table 4 maps to the layout shown in Figure 1.

Figure 1: Device View of Connector

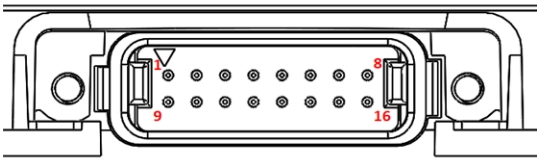
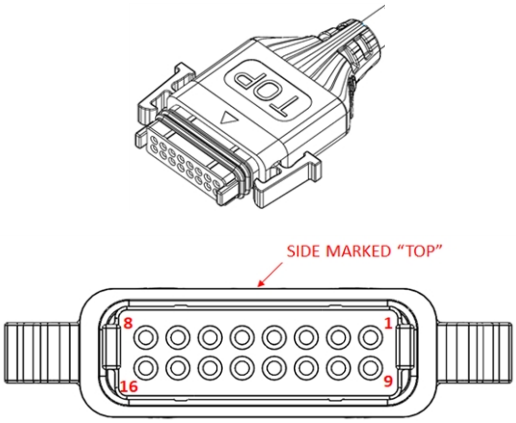


Figure 2: GT 12xx Cable Connector View



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Table 4: Electrical Pin Assignment

PIN	Function	Comment
1	9/12 V Power Out	-
2	I/O_1	See Note below
3	I/O_2	See Note below
4	I/O_3	See Note below
5	I/O_4	See Note below
6	CAN L / RS-485 B	-
7	RS-232 Tx	-
8	HV Analog 1	-
9	Ground	-
10	Ground	-
11	Ground	-
12	External Power In	-
13	1-Wire Bus	-
14	CAN H / RS-485 A	-
15	RS-232 Rx	-
16	HV Analog 2	-

Note: The GT 12xx provides four (4) general purpose I/Os that support a variety of wired and wireless sensors.

The following I/O configurations are possible, for a total of four (4): four (4) wired I/Os, three (3) wireless I/Os, and only one (1) wireless door sensor.

With the exception of an I/O supporting a wireless door sensor, no additional logic is associated with wired or wireless I/Os.

If using a wired door sensor, it can only be configured as a binary event (for example, Open/Closed).

If using an I/O to support the wireless door sensor, it provides additional logic. For GT 1210 / GT 1230 devices with integrated cargo sensors, this wireless door sensor triggers a cargo scan when a door state change event occurs.

2.4 I/O Interface

2.4.1 Configurable I/Os

The device supports four (4) configurable I/Os (I/O_1 to I/O_4) that can operate in one of the following modes:

- Digital input with weak (1 MΩ) pull-down
- Digital input with 30-50 K pull-down (processor internal pull-down)
- Digital input with 30-50 K pull-up (processor internal pull-up)
- Analog input with 0 to 3.3 V range
- Digital output – push-pull
- Digital output – open drain switch-to-ground

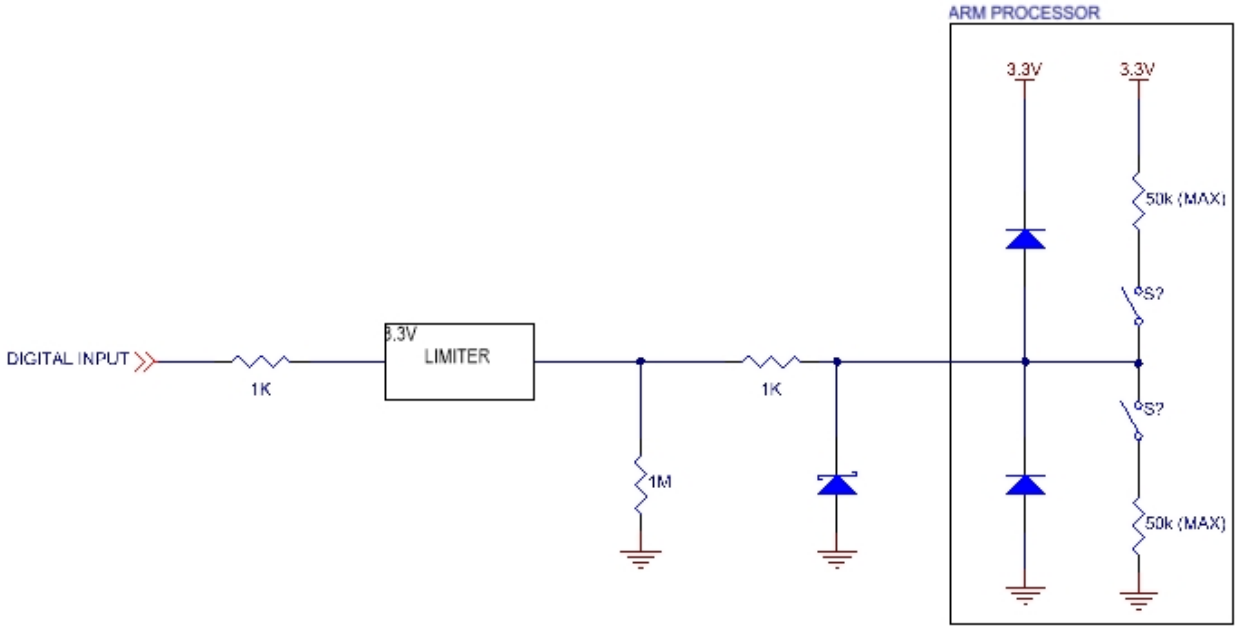
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- Thermistor input
- Disabled

2.4.1.1 Digital Input

Figure 3 shows a schematic of the I/O when configured as a digital input.

Figure 3: Digital Input



Input Type	S1	S2
With pull-down	Open	Closed
With pull-down	Closed	Open
With weak pull-down	Open	Open

The input specifications are provided in the table below.

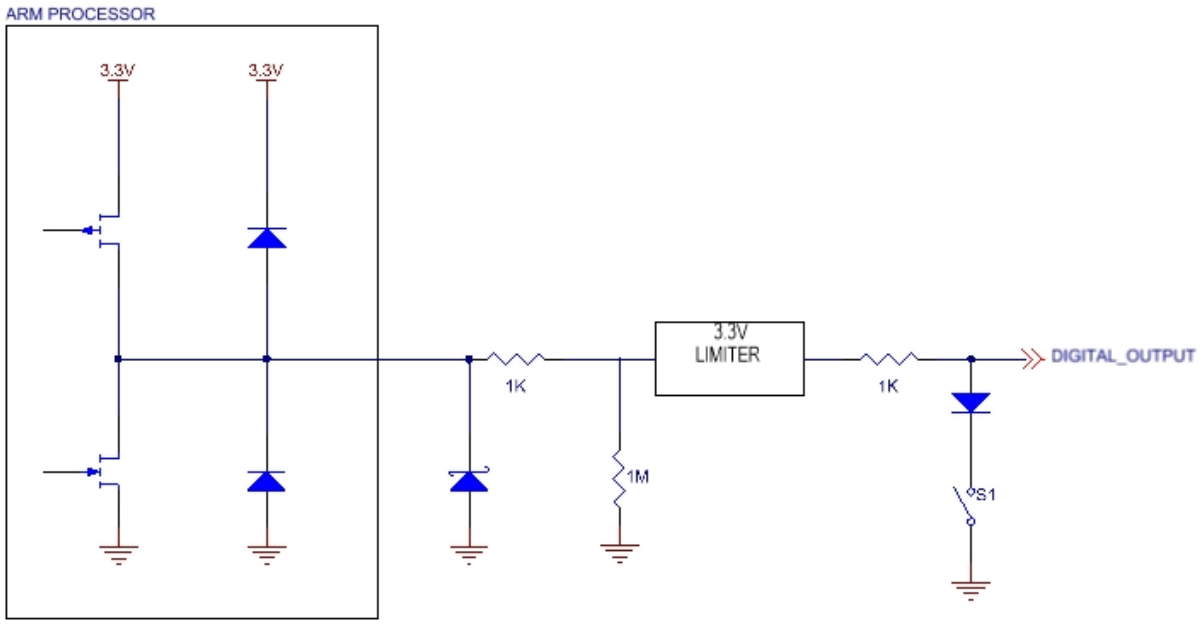
Parameter	Min.	Typical	Max.	Units
Input low range	-10	-	1.15	V
Input high range	2.30	-	200	V
Input current, pull-up/pull-down disabled (weak 1 MΩ pull-down still in place); V_{in} = 3.0 V	-	3	-	µA
Input source current with pull-up (V_{in} = 0.0 V)	-	75	-	µA
Input sink current with pull-down (V_{in} = 3 to 200 V)	-	81	-	µA
Input bandwidth	1	-	-	kHz

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2.4.1.2 Digital Output

Figure 4 shows a schematic of the I/O when configured as a digital output.

Figure 4: Digital Output



Push-pull	S1 = Open
Open drain	S1 = Closed (Low Impedance) S1 = Open (High Impedance)

2.4.1.2.1 Push-pull

In the push-pull configuration the output is driven directly from the microprocessor.

Parameter	Min.	Typical	Max.	Units
Output high voltage - open circuit	2.80	-	-	V
Output high voltage (sourcing 25 µA)	2.80	-	-	V
Output low voltage (sinking 25 µA)	-	-	0.5	V
Output bandwidth	100	-	-	Hz

2.4.1.2.2 Switch to Ground

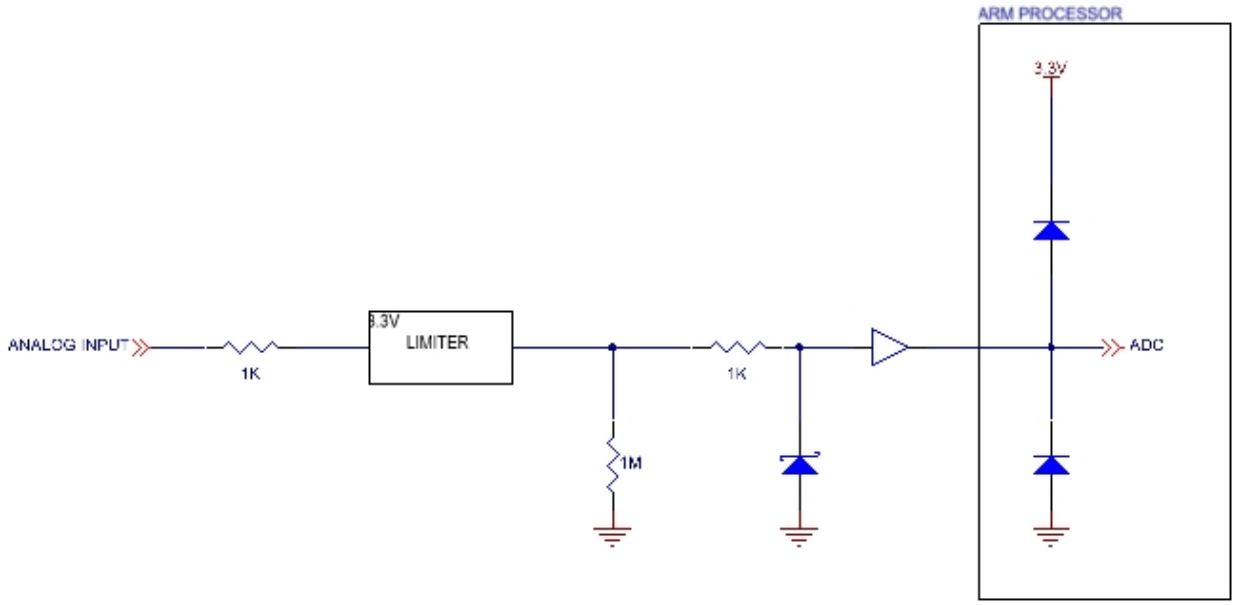
Parameter	Min.	Typical	Max.	Units
Sink current (do not exceed)	-	-	250	mA
Output voltage	-	1.15	1.35	V
Voltage limits (high impedance)	-10	-	150	V
Output bandwidth	100	-	-	Hz

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2.4.1.3 Analog Input

Figure 5 shows a schematic of the I/O when configured as an analog input.

Figure 5: Analog Input



Parameter	Min.	Typical	Max.	Units
Input impedance	-	1	-	MΩ
Minimum measurement voltage	-	0	-	V
Maximum measurement voltage	3.0	3.0	-	V
Resolution (12 bits)	-	0.7	-	mV
Proportional measurement error	-	-	1.5	%
Offset error	-	-	2	LSB
Voltage limits	-10	-	200	V

2.4.1.4 Thermistor Input

Parameter	Min.	Typical	Max.	Units
Pull-up resistance	-	7.5	-	kΩ
Pull-up voltage	-	3.3	-	V

2.4.2 High Voltage Analog Inputs

Examples for the high voltage analog input based on existing sensors and peripherals include:

- Fuel sensor analog input
- Cargo sensor analog input
- Current sense input

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The device includes two dedicated analog inputs that can convert inputs of up to 12 V.

Parameter	Min.	Typ..	Max.	Units
Minimum measurement voltage	-	-	0	V
Maximum measurement voltage	12	-	-	V
Resolution (16 bits)	-	0.2	-	mV
Proportional measurement error	-	-	1.5	%
Offset error	-	-	2	LSB
Voltage limits	-10	-	200	V

2.4.3 Dedicated Output

The device has a dedicated output pin to provide high voltage output or power to external systems such as sensors and external communication channels. This dedicated output is a current limited supply that is protected from any transients and reverse voltage.

The dedicated output voltage is software selectable to either regulated 9 V or 1 2V. Global GT 12xx models can also be set to a direct battery connection.

The direct battery connection allows for low power devices with internal regulation that require continuous power to operate.

Parameter	Min.	Typ.	Max.	Units
Voltage (enabled)	-	7.6 (nom), 9, or 12	-	V
Voltage (disabled)	-	0	-	V
Current output	-	-	1000	mA

Examples for the dedicated output based on existing sensors and peripherals include:

- Fuel sensor power
- Cargo sensor power
- ST 21xx[1] (for direct battery output)

2.5 Serial Interfaces

Devices have the following non-programmable interfaces:

- CAN Bus or RS-485
- RS-232
- 1-Wire

[1]ST 21xx refers to all ORBCOMM products beginning the "ST 21", unless noted otherwise.

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2.5.1 CAN Bus

The device provides the option of either a CAN Bus or RS-485 interface.

The device incorporates a controller area network interface with signaling rates up to 1 Mbps.

Note: You must provide a termination resistor externally to the device.

Parameter	Min.	Typical	Max.	Units
Differential Output Voltage (dominant)	1.2	-	3	V
Differential Output Voltage (recessive) No Load	-0.2	-	0.2	V
CANH or CANL	-35	-	35	V
Voltage Tolerance	-60	-	60	V

2.5.2 RS-485/J1708/Modbus

The device provides the option of either a CAN Bus or RS-485 interface.

Note: The device does not incorporate a termination resistor.

The electrical characteristics of the interface are:

Parameter	Min.	Typical	Max.	Units
Input Common Mode Voltage	-7	-	12	V
Differential Input Threshold	-200	-125	-50	mV
Output Common Mode Voltage	-	1.5	3	V
Differential Drive Output, 54 Ω load	1.5	-	-	V
Voltage Tolerance	-60	-	60	V

2.5.3 RS-232 (Main)

The main RS-232 interface defaults to the following settings: 9600 bit/s, 1 start, 8 data, 1 stop bit, and no parity. The baud rate is configurable up to 115,200 bps.

The RS-232 port allows software upgrades.

The electrical characteristics of the interface are:

Parameter	Min.	Typical	Max.	Units
Rx Input Low Threshold for DTE Connected	-	-	-2.7	V
Rx Input High Threshold for DTE Connected	2.7	-	-	V
Rx Threshold for DTE Disconnected	-0.3	-	0.3	V
Serial Rx Input Low Threshold	0.6	-	-	V
Serial Rx Input High Threshold	-	-	2.4	V
Serial Tx Low Output (3 kΩ load)	-	5.4	-	V
Serial Tx High Output (3 kΩ load)	-	5.4	-	V

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2.5.4 1-Wire

The 1-Wire interface allows connection to downstream 1-Wire devices connected on a bus, or to a single button reader. Relative to any attached 1-Wire device, the transceiver behaves as the master. The 1-Wire driver supports 3 V devices on the bus.

At standard speed, the 1-Wire supports up to 39 devices over a 61-meter (200 feet) CAT5 cable. In overdrive, the usable expected distance is reduced to ≤ 15 meters (≤ 50 feet) with a maximum node count of 9.

The electrical characteristics of the interface are:

Parameter	Min.	Typical	Max.	Units
1-Wire Input High Voltage	1.8	-	-	V
1-Wire Input Low Voltage	-	-	0.6	V
1-Wire Output Low Voltage	-	-	0.2	V
Load Current @ 3 V	-	4	-	mA

2.6 RF Specifications

2.6.1 Satellite Performance

The device complies with the ORBCOMM Subscriber Communicator Specifications.

2.6.2 Cellular Performance

The device uses a PTCRB certified cellular module.

2.6.3 GPS Performance

The GPS has an un-aided cold start of less than 35 seconds and an un-aided hot start in less than 1 second.

The 2D position accuracy is less than 10 m CEP.

2.7 Battery

The GT 12xx is internally powered by a rechargeable long-life, industrial grade 7.4 V nominal battery pack able to supply the current necessary for all operational modes. The battery is charged by the internal solar panel and, optionally, by an external power supply. The battery pack includes a protection circuit with internal NTC sensor that provides all required UN 38.3 protection, including over discharge, overcharge, and over current protections.

The electrical specifications of the battery pack are below.

Parameter	Min.	Typical	Max.	Units
Capacity	5000	-	-	mAh
Voltage (nominal)	-	7.4	-	V
Current Output	3	-	-	A
Over Discharge Limit	-	5.5	-	V
Temperature	-40 (-40)	-	70 (158)	°C (°F)
Battery Chemistry	Lithium-ion Cell			

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2.7.1 Battery Status Measurement

The GT 12xx can measure the voltage of the external power supply and the internal battery independently.

If the device detects a battery voltage below 7.0 V at power on, it enters battery charging mode. In this mode it will not communicate over-the-air or over the serial port. Every hour the device measures the voltage. If the battery voltage measures above 7.0, the device continues to boot up. The device also exits the battery charging mode after 24 hours, regardless of the battery voltage.

The device enters hibernation mode when a voltage below 6.5 V is detected. When in hibernation the battery voltage is checked every hour, until the battery voltage reaches 6.7 V. During hibernation the device stops location updates, cargo scans, and heartbeats.

The device has two methods it can use to measure the battery status:

- Battery voltage measurement
- Battery capacity measurement through a fuel gauge.

2.7.1.1 Battery Voltage Measurement

The device can measure its main input supply voltage over the range 5 to 8.4 V, to an accuracy of ±55 mV.

2.7.1.2 Battery Capacity Measurement

The device includes a fuel gauge capable of calculating the available battery capacity by measuring voltage, current, and temperature. There is no significant impact on the system average current while the device is monitoring battery status.

The key fuel gauge specifications are listed below.

Parameter	Minimum Specification
Average Current	<10 µA
Accuracy	<10 %
Interface	Serial

2.7.2 Battery Controller

Battery hardware monitors and controls the charge and discharge of the battery pack. Discharge above the battery pack maximum discharge temperature is restricted. Charging is limited to between the minimum and maximum charging temperature of the battery pack. Temperature is monitored using the thermistor internal to the battery pack.

The microcontroller is capable of disabling the battery charger to reduce load on the input power source.

The battery is charged by the internal solar panel and, optionally, by an external power supply. A cable is required to charge the battery pack when using an external power supply. Charging of the battery is disabled during cellular communication.

The battery controller electrical specifications are below:

Parameter	Min.	Typical	Max.	Units
External Charge Voltage	10.5	-	32	V
Charge Current	-	-	400	mA
Float Voltage	8.35	-	8.4	V
Charge Time	-	-	12	Hours

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2.7.3 Battery Life Expectancy

Battery life expectancy:

- Devices are charged to a minimum of 7.5 V or better prior to shipping.
- Energy usage varies greatly depending on environmental conditions, how often the device reports, how much information is reported, storage conditions, and other variables. If for example, the device is kept in typical outdoor conditions for container tracking, it can report the GPS location every 5-minutes with little risk of the battery ever draining to a point of losing its charge.
- A fully depleted battery returns to normal operation after 4-6 hours of full sun.
- A fully depleted battery is fully charged after 25 hours of full sun.
- A fully depleted battery returns to normal operation after 2-3 hours of external power.
- A fully depleted battery is fully charged after 12 hours of external power.
- A fully charged GT 12xx can remain in inventory over 1 year without depleting its battery.
- One hour of sunlight provides enough energy for over 100 reports.

2.8 GNSS

The manufacturer's specifications are given in the table below.

Table 5: Multi-GNSS Specifications

Parameter	GPS	GLONASS	BeiDou
Time to First Fix[1]			
Cold Start	29 s	30 s	30 s
Warm Start	28 s	25 s	28 s
Hot Start	1 s	1 s	1 s
Sensitivity			
Tracking	-162 dBm	-158 dBm	-162 dBm
Hot Start	-156 dBm	-156 dBm	-156 dBm
Cold Start	-148 dBm	-140 dBm	-147 dBm
Accuracy			
Horizontal Position (CEP) [2]	2.5/2.0 m	4.0 m	-
Velocity	0.1 m/s	0.1 m/s	-
Heading	0.5 degrees	0.5 degrees	-

[1] All satellites at -130 dBm

[2] CEP, 50%, 24 hours static, -130 dBm

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2.9 Memory

Parameter	Value
EEPROM	2 KB
Main Processor Flash	16 MB, 100,000 write-erase cycles over its operating life.
BLE Flash	2 MB, 100,000 write-erase cycles over its operating life.

2.10 Sensors

Refer to APPENDIX B for a list of GT 12xx supported ORBCOMM sensors and peripherals.

2.10.1 Temperature Sensor

The temperature sensor is off between readings to conserve power.

Parameter	Value
Range	-40°C to +85°C (-40°F to +185°F)
Accuracy (typical)	2°C (3.6°F)

2.10.2 Accelerometer

The accelerometer is important in low power applications where it is critical to save power while the asset is stationary.

The table below shows the minimum accelerometer specifications.

Parameter	Condition	Min.	Typ.	Max.	Units
Acceleration Range	software selectable	-	+2	-	g
		-	+4	-	g
		-	+8	-	g
		-	+16	-	g
Bandwidth Filtering	selectable via digital interface	40	-	400	Hz
Sensitivity	2 g	-	256	-	LSB/g
	4 g	-	128	-	LSB/g
	8 g	-	64	-	LSB/g
	16 g	-	21	-	LSB/g

2.10.3 Magnet Switch

All GT 12xx devices offer a magnet switch. The magnet-activated switch inside the device allows you to physically restart it for diagnostic testing and wake it from shipping mode.

2.10.3.1 Activate or Restart the Magnet

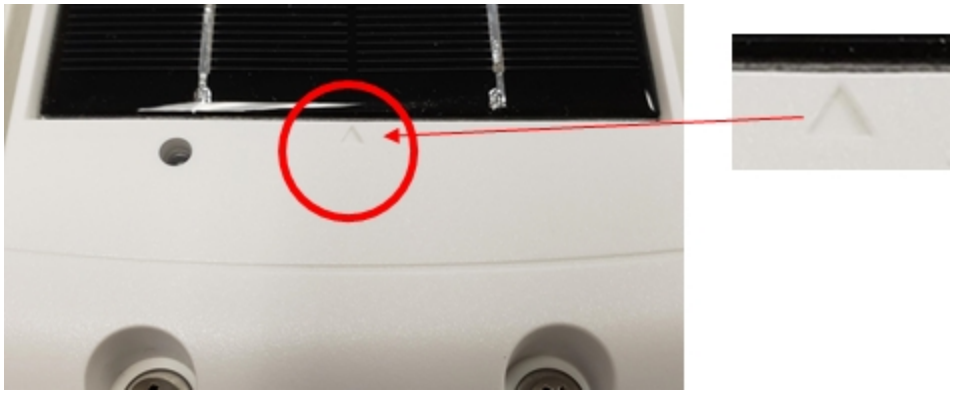
To activate the magnet during initial installation, remove the red label, and then remove the magnet located under the label.

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To restart the magnet:

1. Place the magnet on or slightly raised above the arrow/triangle icon shown in the figure below, hold in place until you see the LED blink blue, about 55-65 seconds, and then remove it. The LED will be solid blue for about 10 seconds while the device restarts.



Place the magnet on or slightly raised above the arrow / triangle icon for	Action	LED activity when the magnet is removed	LED activity when the action is occurring
3 to 7 seconds and then remove	Executes a cargo scan, GNSS location update, and sends a from-terminal message if the asset is moving	Blinks green	Solid green for 15 seconds
55 to 65 seconds and then remove [1]	Causes a restart	Blinks blue	Solid blue for 10 seconds

2.10.4 Tamper Detection

The device offers tamper detection to detect removal from the asset. The magnetic switch reads the presence of a magnet that is lightly adhered to the enclosure and/or mounting bracket. The magnet includes VHB tape that is permanently adheres it to the asset when the device is installed. If the device is removed from the asset, the magnet remains behind, and an alert is generated. Refer to installation guide GT0021, available from ORBCOMM Customer Care or the Technical Publications download website, for more details.

[1] This action only works if the GT 12xx is running firmware v14.1 or higher.

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2.10.5 Cargo Sensor

The GT 1210 / GT 1230 includes a sensor for detecting standard container and trailer loads within an asset

The cargo sensor characteristics are below.

Parameter	Min.	Typical	Max.	Units
Minimum Sensing Distance	-	-	13 (0.5)	mm (inches)
Maximum Sensing Distance	12 (40)	-	-	meters (feet)

The GT 1210 / GT 1230 provides an optional accessory to facilitate high mounting scenarios where the standard zero angle cargo sensor would miss unstacked loads. The high mount accessory includes two cargo sensors placed angles to detect low loads.

2.11 Cellular Module

The device includes a module capable of global cellular communications over LTE, 2G, and 3G networks. The cellular modem is only powered on, when the device has messages to deliver. The modem is powered down after a configurable amount of idle time (15 seconds by default), and the device can not receive messages when the modem is powered down.

The cellular module specifications are below:

Parameter	Minimum Specification
LTE Category	Category 4
LTE Bands	1, 2, 3, 4, 5, 7, 8, 12, 13, 18, 19, 20, 25, 26, 28, 38, 39, 40, and 41
UMTS/HSPA+ Bands	1, 2, 4, 5, 6, 8, and 19
GSM Bands	2, 3, 5, and 8

2.12 BLE Module

The BLE (Bluetooth Low Energy) Module is a low power short range wireless protocol using the 2.4 GHz frequency band. BLE is not compatible with standard Bluetooth.

The BLE allows the device to communicate as a peripheral with a mobile phone or other BLE enabled host for configuration and some debug logging. It allows the device to communicate as a host to local wireless sensors. The BLE system is capable of operating in both the host and peripheral modes at the same time.

The BLE specifications are provided in the table below.

Parameter	Value.	Units
Frequency	2360 min 2500 max	MHz
BLE Version Compliance	5.0	-
Receive Sensitivity	96	dBm
Transmit Power	8	dBm
Available RAM	256	kBytes
Peripherals Available	4	-

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2.12.1 GT 1200 / GT 1220 BLE RF Interface

The GT 1200 / GT 1220 BLE includes a provision for both internal and external antennas that are software selected by the BLE controller. The internal antenna is internal to the GT 1200 / GT 1220 enclosure. The external antenna is cabled to an antenna placed remotely from the GT 1200 / GT 1220.

2.12.2 GT 1210 / GT 1230 BLE RF Interface

The GT 1210 / GT 1230 BLE includes a provision for two antennas internal to the GT 1210 / GT 1230 enclosure selectable by the BLE controller. The purpose of the two antennas is to communicate with BLE devices both inside and outside of the asset. The inside asset antenna utilizes the built-in cargo sensor to communicate with BLE devices such as door sensors and temperature sensors. The outside asset antenna is omnidirectional to communicate with BLE devices outside the asset, such mobile phones with the Field Support Tool.

2.13 LED

The GT 12xx devices include a single LED indicator located on the top of the enclosure.

The LED indicator is under software control but typically indicates general device health during initial power on, otherwise the LED is off.

Green LED

The green LED is solid on for 15 seconds in the following cases:

- When the device starts the application, including activation from shipping mode
- When I/O_1 becomes high

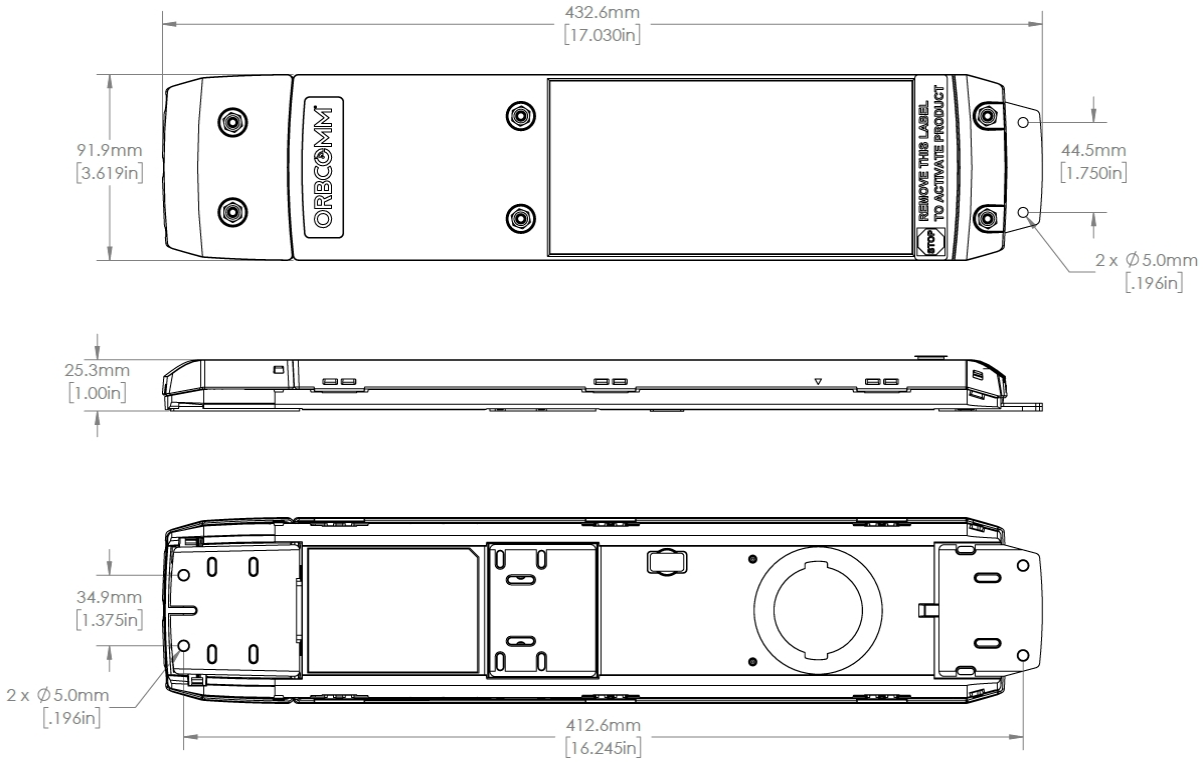
Blue LED

- The Blue LED turns on solid, while in the bootloader. The device enters the bootloader after a restart.

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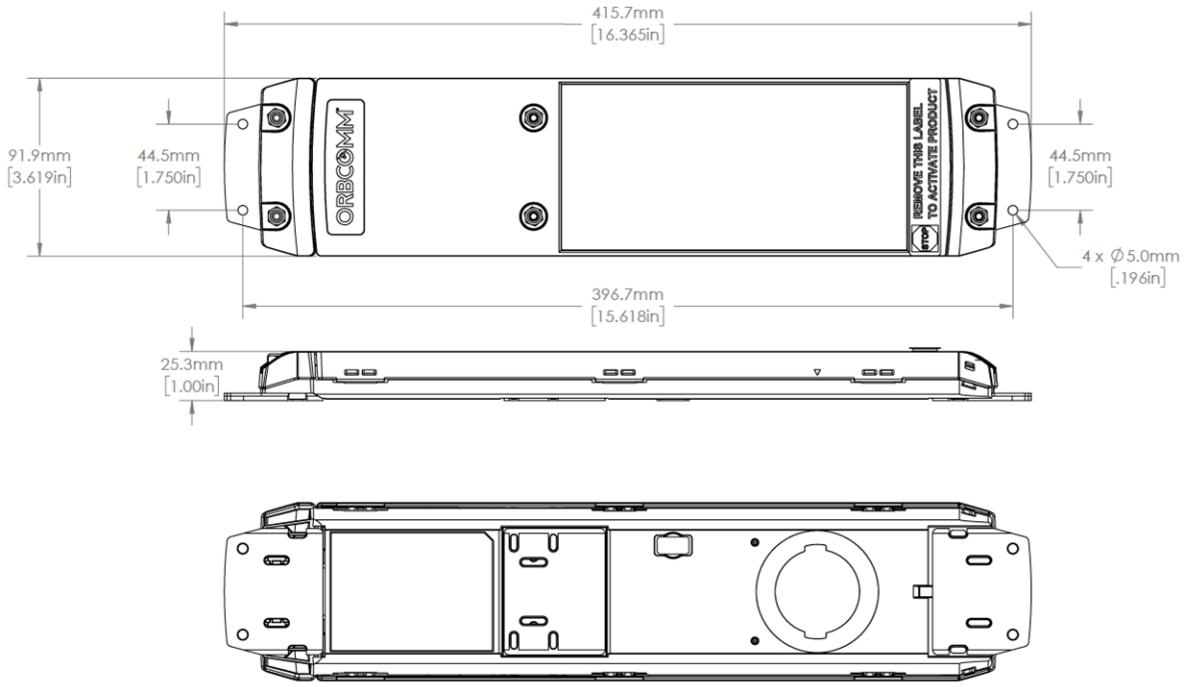
2.14 Mechanical

Figure 6: GT 1200 / GT 1220 with Connector Dimensions



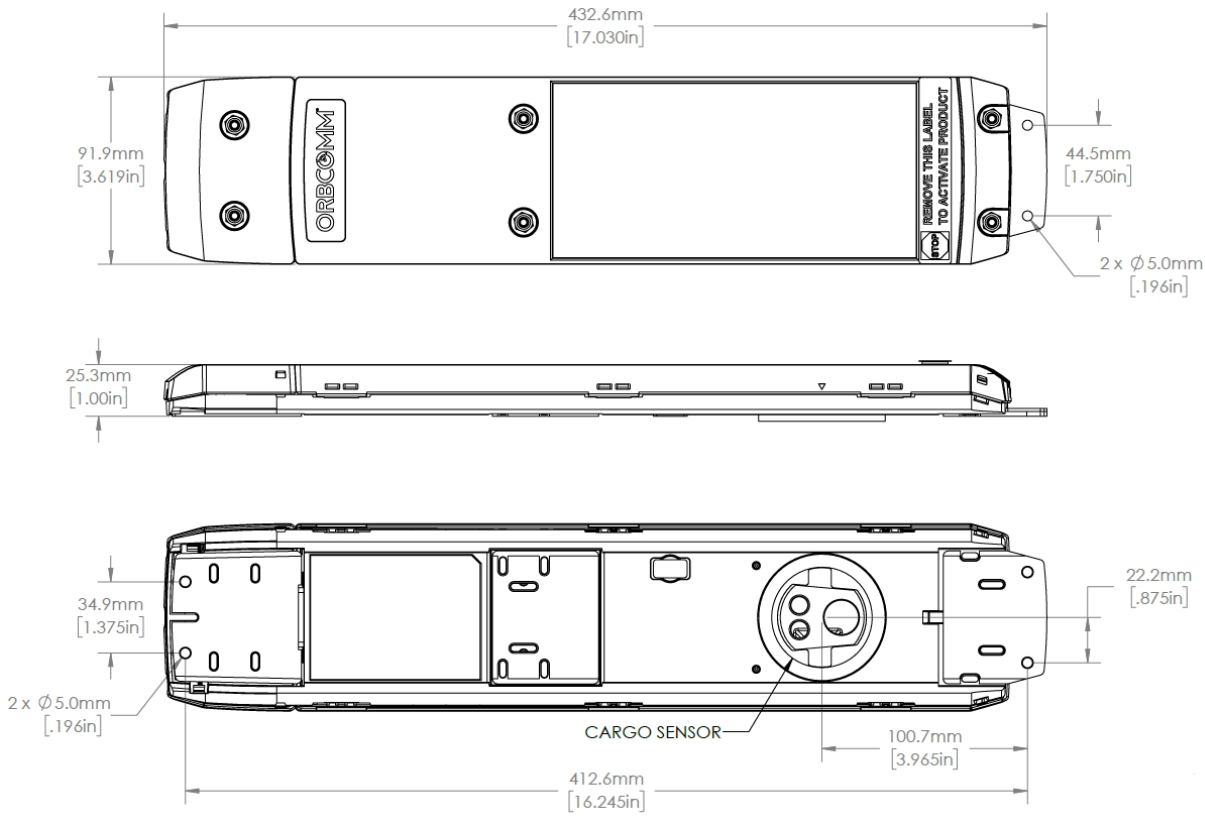
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Figure 7: GT 1200 / GT 1220 without Connector Dimensions



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Figure 8: GT 1210 / GT 1230 with Connector



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Figure 9: GT 1210 / GT 1230 without Connector

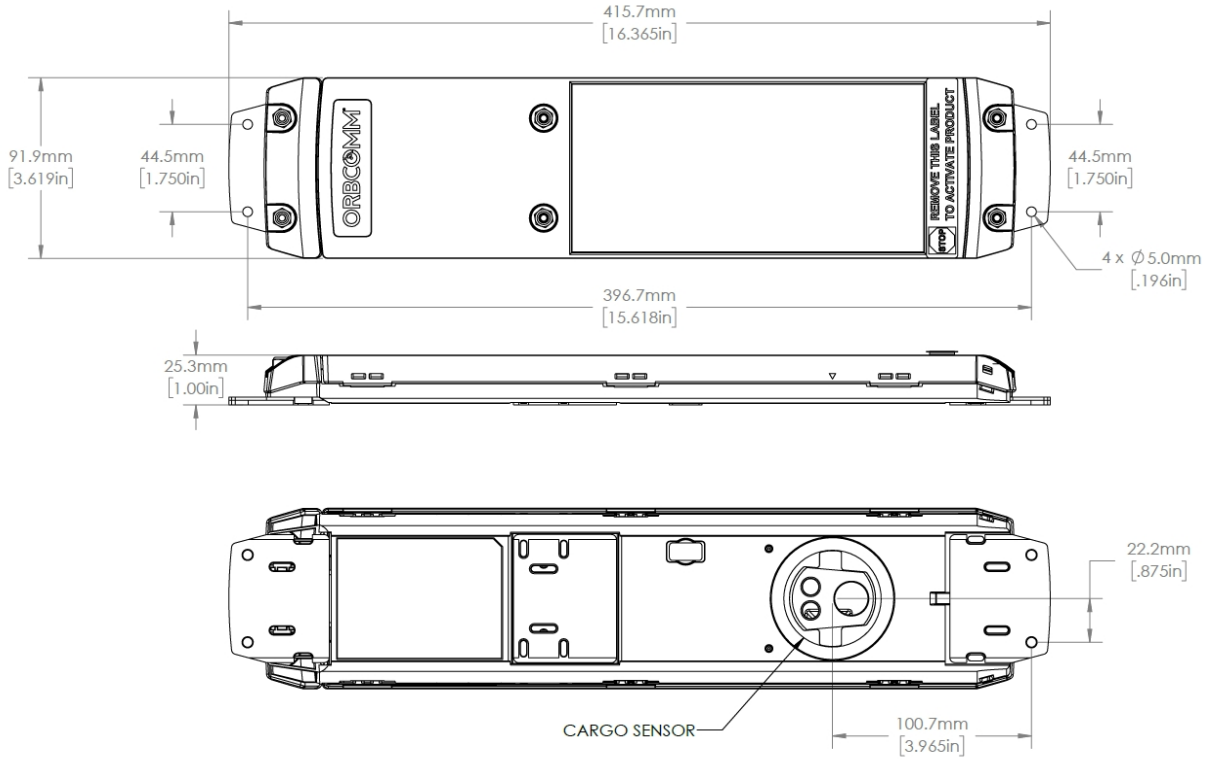


Figure 10: GT 1210 / GT 1230 with Integrated Cargo Sensor and Tamper Detection Seal (underside)



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2.15 Environmental

Parameter	Description
Humidity	The device meets all its specifications during exposure to 90% relative humidity at +70°C (158°F), per the test methodology of SAE J1455, section 4.2.3 (8-hour humidity cycle per figure 4a)
Vibration	The device meets all its specifications during exposure to random vehicular vibration levels per SAE J1455, section 4.9.4.2 figures 6, 7, and 8, and MIL-STD-810G, section 514.6, figure 514.6C-1.
Mechanical Shock	The device meets all its specifications after exposure to positive and negative saw tooth shock pulses with peaks of 20 G and durations of 11 ms as specified in MIL-STD-810G, section 516.6, Procedure I, section 2.3.2c.
Thermal Shock	The device meets all of its specifications after a thermal shock test as detailed in SAE J1455, section 4.1.3.2
Drop Test	The device meets all its specifications after a handling drop test as specified in SAE J1455, section 4.11.3.1.
Altitude	The device meets all specifications after a nonoperating 12.2 km (7.5 miles) altitude test as detailed in SAE J1455, section 4.9.3, except with an ambient temperature of -40°C (-40°F).
Salt Spray Atmosphere	The device meets all of its specifications after a salt spray test as detailed in SAE J1455, section 4.3.3.1.
Immersion	The device meets all of its specifications after a 6-hour alternating hot/cold salt water immersion test as detailed in SAE J1455, section 4.3.3.2. These immersions exclude the cable mating with the external connector.

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Parameter	Description
Exposure to Chemicals and Oils	The device meets all of its specifications after a light to moderate splash test as detailed in SAE J1455 section 4.4.3.2, for the following chemicals: Window Washer Solvent Gasoline Diesel Fuel Fuel Additives Alcohol Antifreeze Water Mixture Degreasers Soap and Detergents Steam Waxes Kerosene Freon Spray Paint Paint Strippers Ether Dust Control Agents (magnesium chloride) Moisture Control Agents (calcium chloride) Ammonia Aluminum brightener (acid wash)
Steam Cleaning and Pressure Washing	The device meets all of its specifications after a steam cleaning and pressure wash test as detailed in SAE J1455, section 4.5.3.
Fungus	The device meets all of its specifications after a fungus test as detailed in SAE J1455, section 4.6.3.
Dust and Sand Bombardment	The device meets all of its specifications after a dust and sand bombardment test as detailed in SAE J1455, section 4.7.3.
ESD	All electrical interfaces operate normally after being subjected to 8 kV ESD contact discharge per IEC 61000-4-2 human body model, level 3. (Enclosure) All electrical interfaces operate normally after being subjected to 6 kV ESD contact discharge per IEC 61000-4-2 human body model, level 3.

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3 COMPLIANCE

The GT 12xx devices have obtained the following certifications:

FCC

- CFR Title 47: Telecommunication, Part 15 - Radio Frequency Devices

Industry Canada

- ICES-003

PTCRB

- GT 1210 / GT 1230 certified

RoHS

- Restriction of Hazardous Substances (RoHS)[1]

Ingress Protection

- IP67

Brazil and South Africa

EU Declaration of Conformity

Hereby, ORBCOMM Inc. declares that the radio equipment types listed in this document comply with Directive 2014/53/EU.

The full text of the EU declaration of conformity is available from http://www2.orbcomm.com/eudoc.

[1]European Union's (EU) Directive 2015/863/EU "Restriction of Hazardous Substances" (RoHS) in Electronic and Electrical Equipment.

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APPENDIX A ORDER PART NUMBER

GT Devices	Part Number
GT 1200 with 16-Pin Connector (LTE/3G/2G)	GT1200-1100-C
GT 1200 without Connector (LTE/3G/2G)	GT1200-0100-C
GT 1200 with 16-Pin Connector (Global LTE)	GT1200-1700-S
GT 1200 without Connector (Global LTE)	GT1200-0700-S
GT 1210 with 16-Pin Connector (LTE/3G/2G, integrated Cargo Sensor)	GT1210-1100-C
GT 1210 without Connector (LTE/3G/2G, integrated Cargo Sensor)	GT1210-0100-C
GT 1210 with 16-Pin Connector (Global LTE, integrated Cargo Sensor)	GT1210-1700-S
GT 1210 without Connector (Global LTE, integrated Cargo Sensor)	GT1210-0700-S
GT 1220 with 16-Pin Connector (US only, LTE/3G/2G)	GT1220-1101-C
GT 1220 without Connector (US only, LTE/3G/2G)	GT1220-0101-C
GT 1220 with 16-Pin Connector (Global LTE)	GT1220-1701-S
GT 1220 without Connector (Global LTE)	GT1220-0701-S
GT 1230 with 16-Pin Connector (US only, LTE/3G/2G, integrated Cargo Sensor)	GT1230-1100-C
GT 1230 without Connector (US only, LTE/3G/2G, integrated Cargo Sensor)	GT1230-0100-C
GT 1230 with 16-Pin Connector (Global LTE, integrated Cargo Sensor)	GT1230-1700-S
GT 1230 without Connector (Global LTE, integrated Cargo Sensor)	GT1230-0700-S
Accessories (GT 12xx device not included)	**Part Number**
GT 12xx Replacement Mounting Kit (re-install kit when moving the device to another asset)	ST101018-001
Temporary Magnet Mount Kit (includes carabiner tether)	ST100904-001
Right-angle Cable Cover (covers the exposed cable into the trailer)	ST101030-001
GT 12xx 16-Pin Blunt-cut Cable (1.2 m / 4 ft)	ST100859-001
Power Cable (external power only, no ring terminals, 6.7 m / 22 ft., also used for charging the device)	ST101026-001
External Power Cable (see APPENDIX D)	ST101025-001
Sheet and Post Horn Kit - 2 1/8" to 2 1/2"	ST100965-001
Sheet and Post Horn Kit - 1 3/4 to 2 1/8"	ST100965-002
Universal Sheet and Post Horn Kit (both horns)	ST100965-003
Sensors - see APPENDIX B	
Contact your Account Executive for additional products and ordering codes	

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APPENDIX B SUPPORTED SENSORS

Supported sensors are for use with GT 12xx devices with 16-pin connectors. Refer to the appropriate data sheet or contact your ORBCOMM Account Manager for a complete list of sensors and part numbers.

Note: Additional extension and/or power cables may be required to connect the sensors to the GT 12xx.

	Product Name	Product Description
	Wireless Door Sensor (DS 300)	The DS 300 Wireless Door Sensor (p/n ST100547-001) can detect if a trailer or container door is open or closed, or while on route, providing enhanced security and operational efficiency. The magnetic door sensor monitors the status of the trailer door. The door sensor features an interrupt-driven digital interface that minimizes power draw from the sensor. The wireless door sensor can be configured over-the-air and integrated with additional sensors.
	Wired Door Sensor	The door sensor (wired) can detect if a trailer or container door is open or closed, or while on route, providing enhanced security and operational efficiency. The magnetic door sensor monitors the status of the trailer door. The door sensor features an interrupt-driven digital interface that minimizes power draw from the sensor.
	Wireless Temperature Sensor (TS 310)	A wireless temperature sensor (ST100643-002) and an optional mounting bracket (ST101451-001). The sensor provides both general cargo area temperature and multi-zone monitoring in refrigerated trailers and trucks, independent of the refrigeration unit.
	Wired Temperature Sensor	ORBCOMM's proven temperature sensors (LK4048 (6 m/20 ft.), LK4049(15 m/49 ft.) or L4050 (22 m/72 ft.) help to ensure the integrity of perishable and temperature-sensitive products. This technology enables the collection and analysis of in-transit cargo temperature data for the entire trip to maximize compliance, help prevent loss of loads and improve customer satisfaction. Specific temperatures or ranges can be set and verified for the duration of the trip. The sensor can also send near-real-time alerts if the temperature deviates from the set range. GT 12xx must have firmware v2.14.3 or higher to support a temperature sensor.

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APPENDIX C GT 12XX BLUNT-CUT CABLE

The GT 12xx 16-Pin blunt-cut cable (p/n ST100859-001) is available in a 1.2 m / 4 ft. cable length.

Figure 11: Blunt-Cut Cable (mm)

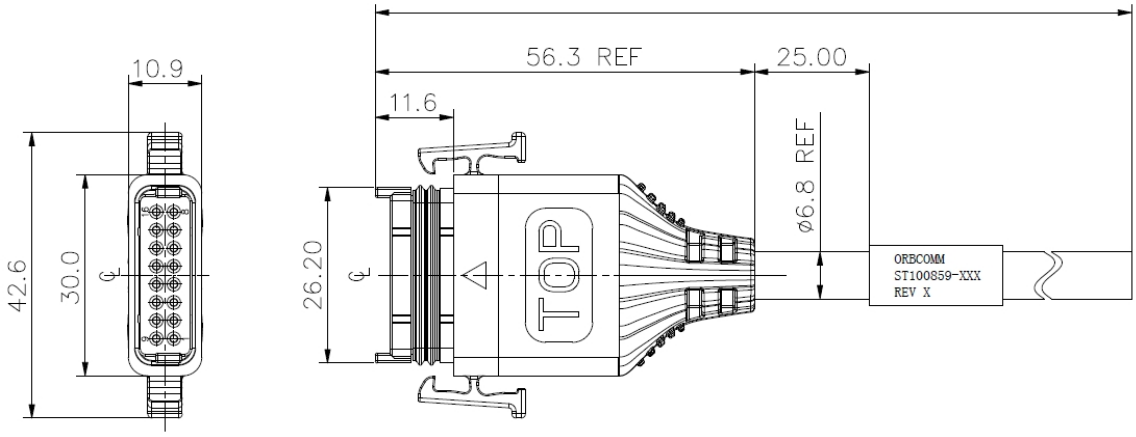
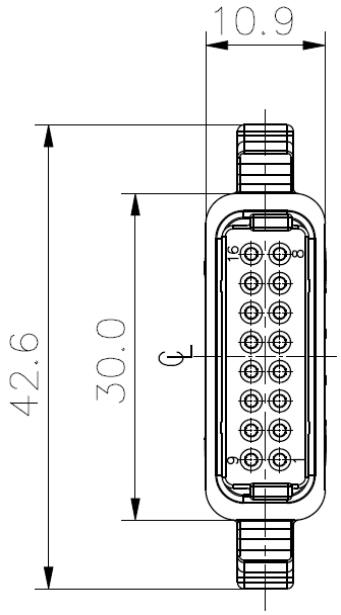


Figure 12: Pin Connector Numbering



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Table 6: Pin Descriptions

Pin #	Signal Name	Color
1	9 V / 12 V PWR	Gray
2	I/O_1	White (see Note below)
3	I/O_2	Orange (see Note below)
4	I/O_3	Purple (see Note below)
5	I/O_4	Green (see Note below)
6	AUX_SERIAL B	White with a trace of Gray
7	RS-232 Tx	Brown
8	HV_Analog 1	Blue with a trace of White
9	GND	Black with a trace of Red
10	GND	Black with a trace of White
11	GND	Black (22 AWG)
12	EXT_PWR	Red (22 AWG)
13	1-Wire_I/O	Blue
14	AUX_SERIAL A	Green with a trace of White
15	RS-232 Rx	Yellow
16	HV_Analog 2	Orange with a trace of White

Note: The GT 12xx provides four (4) general purpose I/Os that support a variety of wired and wireless sensors.

The following I/O configurations are possible, for a total of four (4): four (4) wired I/Os, three (3) wireless I/Os, and only one (1) wireless door sensor.

With the exception of an I/O supporting a wireless door sensor, no additional logic is associated with wired or wireless I/Os.

If using a wired door sensor, it can only be configured as a binary event (for example, Open/Closed).

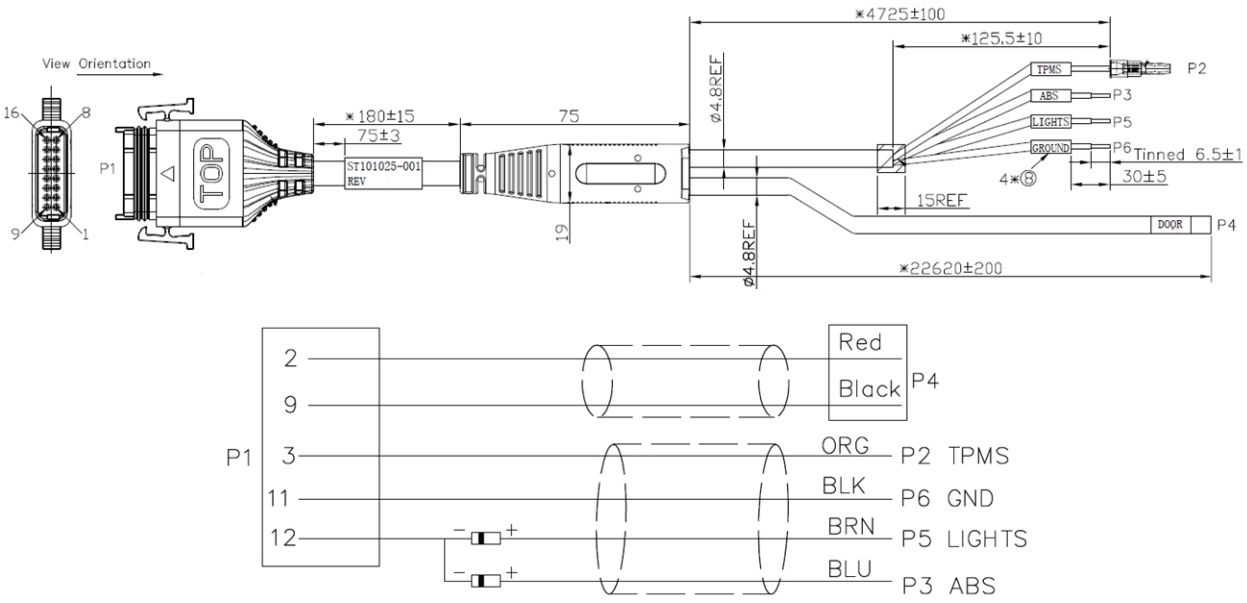
If using an I/O to support the wireless door sensor, it provides additional logic. For GT 1210 / GT 1230 devices with integrated cargo sensors, this wireless door sensor triggers a cargo scan when a door state change event occurs.

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APPENDIX D EXTERNAL POWER CABLE

The External Power Cable (p/n ST101025-001) includes TPMS light, wired door cable, p-clamps, drilling screws, butt splices, and heat shrink

Figure 13: Power Cable with TPMS (mm)



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